FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Advanced)-2020

PAPER –1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. ABD

Sol. $q = \frac{CV}{3}\left(1-e^{-t/RC}\right)$

Current through V_2 $\quad i_2 = \frac{V}{3R}\left(e^{-t/RC}\right)$

Current through V_1 $\quad i_1 = \frac{V}{2R} - \frac{V}{6R}e^{-1/RC}$

at t = RC, power delivered by $V_2 = Vi_2 = \frac{V^2}{3Re}$

2. ABD

Sol. Gravitational field intensity inside the spherical shell is zero.
Gravitational force is always directed centre of spherical mass for outside points.
Force on P act only due to m_1 because point P is outside the spherical shell of mass m_1.
So choices (A), (B) and (D) are correct.

3. BCD

Sol. Apply conservation of mechanical energy to obtain x.

4. BC

Sol. As $\lambda_0 = \frac{hc}{eV}$ and as V increases. λ_0 decreases but characteristic wavelengths do not change so interval between $\lambda_{k\alpha}$ & λ_0 increases and the same for the interval between $\lambda_{k\beta}$ & λ_0

5. BC

Sol. $50 \times 10^{-6} = \dfrac{r \times i}{100 + r} \Rightarrow$ for I = 5 mA

$r = 1\Omega$.

$V = 50 \times 10^{-6} (100 + r_o) = 50 \times 10^{-6} r_o$

6. AB

Sol. Assume unknwon currents and apply KVL twice.

7. BD

Sol. For n^{th} mode of vibration $\lambda_n = \dfrac{4\ell}{(2n-1)}$.

8. B

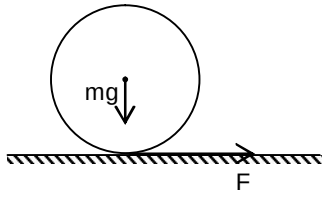
9. B

10. A

Sol. **(for Q.8 -10)**

Linear impulse $\int F dt = mv_0$ …(i)

Angular impulse $\int F h dt = I_c \omega_0$ …(ii)



$\Rightarrow \quad mv_0 h = I_c \omega_0$ and $I_c = \dfrac{2}{5} mR^2$

Force of friction is forward.

$\therefore \quad fR = I_c \alpha \quad \therefore \quad \alpha = \dfrac{\mu m f R}{I_c}$

$a_{cm} = \dfrac{F}{m} = \dfrac{\mu mg}{m} = \mu g$ forward.

11. B

Sol. $k_{effective}$ in (II) is k and $g_{effective} = \dfrac{g}{2}$.

12. C

Sol. $k_{effective}$ in (III) is 4k and $g_{effective} = \dfrac{g}{2}$.

13. A

Sol. $k_{effective}$ in (IV) is $\dfrac{k}{4}$ and $g_{effective} = \dfrac{g}{2}$.

SECTION – B

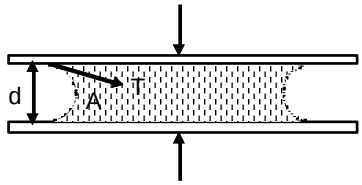
14. 2

Sol. Let R be the radius of the circular layer of water.

Then, $\pi R^2 d \times \rho = m$ …(1)

Pressure at $A = p_0 - \dfrac{2T}{d}$ …(2)

Thus pressure between the plates is less than the atmospheric pressure and so the plates are pressed together as though attracted towards each other.



F, force of attraction = $\Delta p \times$ area $\Rightarrow$ $F = \frac{2T}{d} \times \pi R^2$

$\Rightarrow$ $F = \frac{2T}{d} \times \frac{m}{d \cdot \rho} = \frac{2Tm}{d^2 \cdot \rho} = \frac{2 \times 0.4 \times 10^{-3} \times 0.08}{0.01^2 \times 10^{-4} \times 1000}$ = 6.4 N

15. 6

Sol. As source (horn of bus) is approaching stationary wall (say, listener), therefore, apparent frequency striking the wall is

$$v' = \frac{\upsilon\, v}{\upsilon - \upsilon_s} \qquad \ldots(1)$$

Sound of this frequency will be reflected by the wall (now, source). The passenger is the listener moving towards source. Therefore, frequency heard by the listener

$$v'' = \frac{(\upsilon + \upsilon_L)v'}{\upsilon}$$

using (1)

$$v'' = \frac{\upsilon + \upsilon_L}{\upsilon} \times \frac{\upsilon v}{\upsilon - \upsilon_s} = \frac{(\upsilon + \upsilon_L)v}{\upsilon - \upsilon_s}$$

$$= \frac{(330 + 5) \times 200}{330 - 5} = \frac{335}{325} \times 200$$

$v'' = 206\,\text{Hz}$

$\therefore$ Beat frequency = $(v'' - v)$

$= 206 - 200 = 6\,\text{Hz}$

16. 9

Sol. $a = \frac{21}{14}$, $T_B = 6a = \frac{6 \times 21}{14} = 9$ N.

17. 5

Sol. As stable equilibrium, U is minimum. Thus $\frac{d^2U}{dx^2} > 0$.

And $\frac{dv}{dx} = 0$.

$= \frac{1}{dx}\left(\frac{x^3}{3} - \frac{ax^2}{2} + 20x\right) = 0.$

$\Rightarrow$ $x^2 - 9x + 20 = 0.$ $\Rightarrow$ $(x - 5)(x - 4) = 0.$

x = 5 and x = 4 are points of equilibrium.

And U minimum when $\frac{d^2U}{dx^2} > 0$. i.e. at x = 5.

18. 3

Sol. $F = \frac{dP}{dt}$ = rate of change of momentum of the photons. Thus, in case of complete absorption, F = Intensity of the wave / speed of light.

Chemistry

PART – II

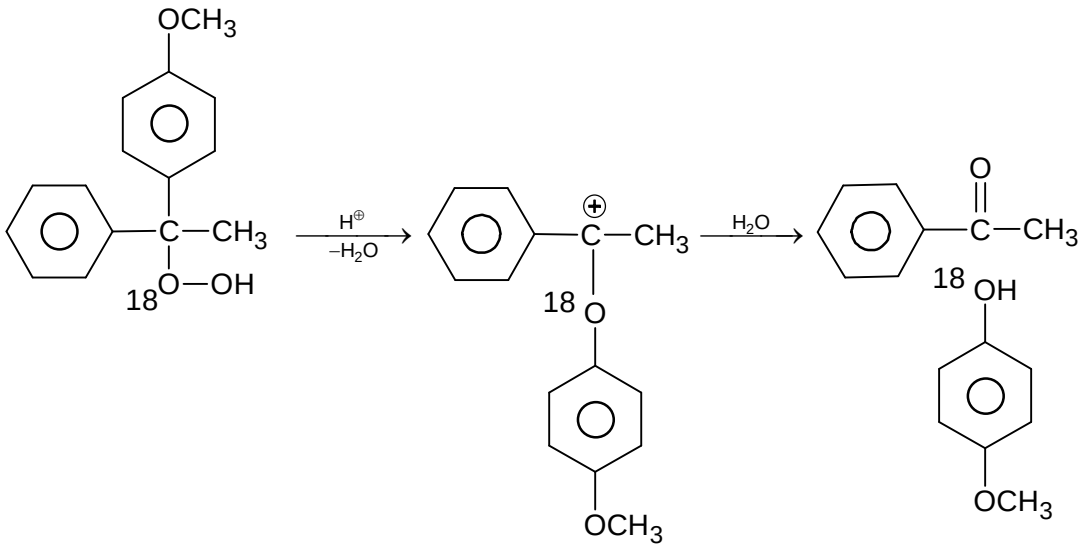
SECTION – A

19. ABC
Sol. (A) EAS, (B) Oxidation & (C) Diel's alder reaction gives Anthraquinone.

20. BC
Sol. Its chemisorption, so heat is released and $O_2 \longrightarrow O_2^-$ electrons in π_{2p}^* increases.

21. BD
Sol. (B) both give N_2
(D) both give NH_3

22. B
Sol.



23. ABD
Sol. Balanced equation is
$6I^- + ClO_3^- + 6H_2SO_4 \longrightarrow Cl^- + 6HSO_4^- + 3I_2 + 3H_2O$

24. ABC
Sol. In (A) angle strain is relieved. B & C have strong electron withdrawing groups, so forms hydrates.

25. AB
Sol. (A) Ortho effect makes O – Me $C_6H_4NH_2$ weakest base. C_6H_5NHMe (PK_a 4.84) P – Me – $C_6H_4NH_2$(Pka 5.10)
(B) O – NO_2 $C_6H_4NH_2$ (Steric effect and H - bonding)
O – Me $C_6H_4NH_2$(PK_a 4.38) < O - MeO $C_6H_4NH_2$(PK_a 4.49) < O – OH $C_6H_4NH_2$(PK_a 4.72)

26. A

Sol. $Me_2C(OH)-CMe_2(NH_2) \xrightarrow[0-5^oC]{NaNO_2+HCl}$ Pinacol rearrangement

Involves Carbocation

27. B

Sol. $CH_3COOC_2H_5 \xrightarrow{NaOEt}$ Claisen condensation & involves carbanion

28. B

Sol. Its carbylamine reaction, involving carbene

29. A

Sol. $P_4 + \overset{-}{O}H \longrightarrow PH_3 + H_3\overset{+1}{P}O_2$

30. B

Sol. $\text{Red } P_4 + \overset{-}{O}H \longrightarrow H_4P_2^{+4}O_6$

31. C

Sol. $PCl_3 + H_3PO_3 \longrightarrow H_4P_2^{+3}O_5$

SECTION – B

32. 6

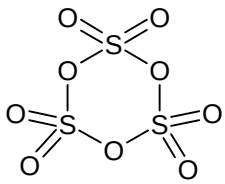
Sol. No. of peptide bonds = 9

Mass of hydrolysed products = $796 + 18 \times 9 = 958$

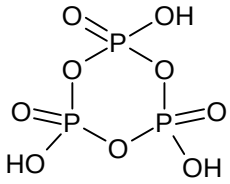
No. of glycine units = $958 \times \frac{47}{100} \times \frac{1}{75} = 6$

33. 0

Sol.

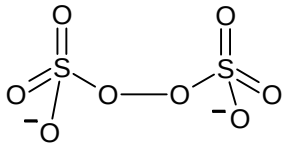


$m_1 = 0$

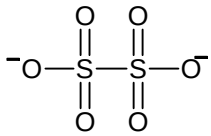


$(HPO_3)_3$

$m_2 = 0$



$m_3 = 1$

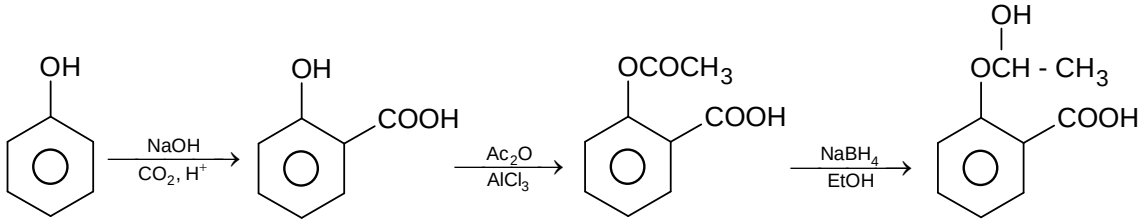


$(S_2O_6^{2-})$

$m_4 = 1$

34. 4

Sol.



Aspirin

35. 2

Sol. $\pi = iCRT$

For monobasic acid $i = (1 + \alpha)$

$2.69 = (1 + \alpha)\ 0.1 \times 0.082 \times 298$

$\Rightarrow \alpha = 0.1$

$|H^+| = C\alpha = 10^{-2}$

pH = 2

36. 8

Sol. Except (viii) all are correct.

Mathematics

PART – III

SECTION – A

37. AC

Sol. The given condition implies

$\cos(2\theta - 2\alpha) + \cos(2\theta + 2\alpha) = 4\sin^2\theta$

$\Rightarrow \quad \cos 2\theta(1 + \cos 2\alpha) = 1$

$\Rightarrow \quad 2\cos^2\alpha = \sec 2\theta \geq 1$

$\Rightarrow \quad |\cos\alpha| \geq \frac{1}{\sqrt{2}}$

38. ABC

Sol. (A) $I = \int_0^{\pi/2} \ln(\cot x)\,dx \Rightarrow I = \int_0^{\pi/2} \ln(\tan x)\,dx$

$I = -\int_0^{\pi/2} \ln(\cot x)\,dx \Rightarrow I = -I \Rightarrow I = 0$

(B) $I = \int_0^{2\pi} \sin^3 x\,dx = -\int_0^{2\pi} \sin^3 x\,dx \Rightarrow I = 0$

(C) $\int_e^{1/e} \frac{-\left(\frac{1}{t^2}\right)dt}{\frac{-1}{t}(\ln t)^{1/3}} = -\int_{1/e}^{e} \frac{dt}{t(\ln t)^{1/3}}$

$I = -I$

$I = 0$

(D) $\sqrt{\frac{1+\cos 2x}{2}} > 0$

$\int_0^{\pi} \sqrt{\frac{1+\cos 2x}{2}}\,dx > 0$

39. ACD

Sol. $(r_1 - r)(r_2 - r)(r_3 - r)$

$= \left(\frac{\Delta}{s-a} - \frac{\Delta}{s}\right)\left(\frac{\Delta}{s-b} - \frac{\Delta}{s}\right)\left(\frac{\Delta}{s-c} - \frac{\Delta}{s}\right)$

$= \frac{\Delta^3 abc}{s(s-a)(s-b)(s-c).s^2} = \frac{\Delta abc}{s^2} \quad = \frac{4abc\Delta}{(a+b+c)^2} = \frac{(abc)^2}{R(a+b+c)^2}$ etc.

40. AB

Sol. As $2x + 3y - 5 = 0$ and $3x - 2y - 1 = 0$ intersect at (1, 1) so this point lies on

$t^2x + ty - 6 = 0$

$\Rightarrow \quad t^2 + t - 6 = 0$

$\Rightarrow \quad (t+3)(t-2)=0$

Hence, $t=-3, 2$

41. ABC

Sol. Let equation of direct common tangent(s) be

$(y-0)=m(x-3)$

$\Rightarrow \quad mx-y-3m=0$

As, $p=r$

$$\Rightarrow \quad \frac{|m(1)-0-3m|}{\sqrt{1+m^2}}=1$$

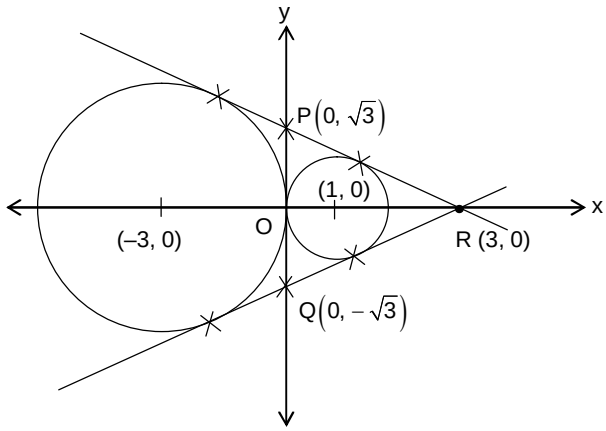
$$\Rightarrow \quad 4m^2=1+m^2$$

$$\Rightarrow \quad m^2=\frac{1}{3}$$

$$\Rightarrow \quad m=\pm\frac{1}{\sqrt{3}}$$

Clearly, ΔPQR is an equilateral triangle.

Now, verify alternatives.



42. ACD

Sol. $\left(9+4\sqrt{5}\right)^n=I+f$

Let $\left(9-4\sqrt{5}\right)^n=f'$

$I+f+f'=\left(9+4\sqrt{5}\right)^n+\left(9-4\sqrt{5}\right)^n$

= Even integer

$0<f+f'<2$

$\therefore$ I is an odd integer

$\Rightarrow (a)$

43. ABCD

Sol. (A) $\lim\limits_{x\to0^+} f(x)=\lim\limits_{x\to0^+}\frac{e^x+a}{2x}=\frac{1}{2}\Rightarrow a=-1$

If $a=-1$, then $\lim\limits_{x\to0^+} f(x)=\frac{1}{2}$, $\lim\limits_{x\to0^-} f(x)=\frac{1}{2}$

$\therefore f(x)$ is continuous at $x=0$, if $a=-1, b=\frac{1}{2}$

(B) $f(x)$ is discontinuous at $x=0$ if $b\neq\frac{1}{2}$.

(C) If $a\neq-1$, then $\lim\limits_{x\to0}\frac{e^x+a}{2x}$ does not exist.

$\therefore x=0$ is a point of irremovable type of discontinuity.

(D) If $a=-1$, then $\lim\limits_{x\to 0} f(x)=\dfrac{1}{2}$

$\therefore b\neq\dfrac{1}{2}\Rightarrow$ removable type of discontinuity at x = 0

44. B

Sol. For parabola $y^2=4ax$, equation of tangent with slope m is $y=mx+\dfrac{a}{m}$ and point of contact $P\left(\dfrac{a}{m^2},\dfrac{2a}{m}\right)$

45. D

Sol. For hyperbola $\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}=1$, equation of tangent with slope m is $y=mx\pm\sqrt{a^2m^2-b^2}$ and point of contact $\left(\pm\dfrac{a^2}{\sqrt{a^2-m^2b^2}},\mp\dfrac{mb^2}{\sqrt{a^2-m^2b^2}}\right)$

46. D

Sol. For circle $x^2+y^2=r^2$, equation of tangent with slope m is $y=mx\pm r\sqrt{1+m^2}$ and point of contact $\left(\mp\dfrac{rm}{\sqrt{1+m^2}},\pm\dfrac{r}{\sqrt{1+m^2}}\right)$

47. B

Sol. $f(x)=\sin^{-1}\left(\dfrac{2x}{1+x^2}\right)=\begin{cases}-\pi-2\tan^{-1}x, & x\le -1\\ 2\tan^{-1}x & -1\le x\le 1\\ \pi-2\tan^{-1}x, & x\ge 1\end{cases}$

$\therefore$ It is non differentiable for $x=\pm 1$

48. A

Sol. $f(x)=\cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)=\begin{cases}-2\tan^{-1}x, & x\le 0\\ 2\tan^{-1}x, & x\ge 0\end{cases}$

It is non differentiable at x = 0

49. A

Sol. $f(x)=\sin^{-1}\left(\dfrac{2x}{1+x^2}\right)=\begin{cases}-\pi-2\tan^{-1}x, & x\le -1\\ 2\tan^{-1}x & -1\le x\le 1\\ \pi-2\tan^{-1}x, & x\ge 1\end{cases}$

It is continuous at x = 0

SECTION – B

50. 5

Sol. $\sin^4\left(\dfrac{\pi}{7}\right)=\dfrac{3}{8}-\dfrac{1}{2}\cos\left(\dfrac{2\pi}{7}\right)+\dfrac{1}{8}\cos\left(\dfrac{4\pi}{7}\right)$ .....(i)

and $\sin^4\left(\frac{3\pi}{7}\right)=\frac{3}{8}-\frac{1}{2}\cos\left(\frac{6\pi}{7}\right)+\frac{1}{8}\cos\left(\frac{12\pi}{7}\right)$

or $\sin^4\left(\frac{3\pi}{7}\right)=\frac{3}{8}+\frac{1}{2}\cos\left(\frac{\pi}{7}\right)-\frac{1}{8}\cos\left(\frac{5\pi}{7}\right)$ .....(ii)

Similarly, $\sin^4\left(\frac{5\pi}{7}\right)=\frac{3}{8}-\frac{1}{2}\cos\left(\frac{10\pi}{7}\right)+\frac{1}{8}\cos\left(\frac{20\pi}{7}\right)$

or $\sin^4\left(\frac{5\pi}{7}\right)=\frac{3}{8}+\frac{1}{2}\cos\left(\frac{3\pi}{7}\right)-\frac{1}{8}\cos\left(\frac{\pi}{7}\right)$ .....(iii)

Equation (i) + (ii) + (iii)

$$\sin^4\left(\frac{\pi}{7}\right)+\sin^4\left(\frac{3\pi}{7}\right)+\sin^4\left(\frac{5\pi}{7}\right)$$

$$=\frac{9}{8}+\frac{1}{2}\cos\left(\frac{5\pi}{7}\right)-\frac{1}{8}\cos\left(\frac{3\pi}{7}\right)+\frac{1}{2}\cos\left(\frac{\pi}{7}\right)$$

$$-\frac{1}{8}\cos\left(\frac{5\pi}{7}\right)+\frac{1}{2}\cos\left(\frac{3\pi}{7}\right)-\frac{1}{8}\cos\left(\frac{\pi}{7}\right)$$

$$=\frac{9}{8}+\frac{3}{8}\left[\cos\left(\frac{\pi}{7}\right)+\cos\left(\frac{3\pi}{7}\right)+\cos\left(\frac{5\pi}{7}\right)\right]$$

$$=\frac{9}{8}+\frac{3}{16}=\frac{21}{16}$$

$\therefore a-b=21-16=5$

51. 4

Sol. $T_1=\frac{1}{\sin 8^o}=\frac{\sin\left(8^o-4^o\right)}{\sin 4^o\sin 8^o}=\cot 4^o-\cot 8^o$ and so on

Given $S=\frac{1}{\sin 8^o}+\frac{1}{\sin 16^o}+\frac{1}{\sin 32^o}+.....+\frac{1}{\sin 4096^o}+\frac{1}{\sin 8192^o}$

$$=\left(\cot 4^o-\cot 8^o\right)+\left(\cot 8^o-\cot 16^o\right)+\left(\cot 16^o-\cot 32^o\right)$$

$$+.....+\left(\cot 2048^o-\cot 4096^o\right)+\left(\cot 4096^o-\cot 8192^o\right)$$

$$=\left(\cot 4^o-\cot 8192^o\right)$$

Also $\cot 8192^o=\cot\left(45\times 180^o+92^o\right)$

$=\cot 92^o$ (As $45\times 180^o=8100^o$)

$\therefore\ S=\cot 4^o-\cot 92^o=\frac{1-\tan^2 2^o}{2\tan 2^o}+\tan 2^o$

$$=\frac{1+\tan^2 2^o}{2\tan 2^o}=\frac{\sec^2 2^o}{2\tan 2^o}=\frac{1}{\sin 4^o}\Rightarrow\alpha=4^o$$

52. 4

Sol. $L: 3^4-\left[{}^3C_1\left(2^4-2\right)+{}^3C_2\right]=36$

M: If $x>0$, $\text{sgn}(x)=1$

$f(x) = 0 + \frac{\pi}{2} = \frac{\pi}{2}$

For $x = 0\, f(x)$ is not defined

$\therefore$ For $x < 0, f(x) = \pi - \frac{\pi}{2} = \frac{\pi}{2}$

$\therefore M = 1$

N : Coefficient of $t^5 =$ coefficient of t^2 in $(1+t^2)^5 \times$ coefficient of t^3 in $(1+t^3)^8$

$= 5 \times 8 = 40$

Hence, L = 36; M = 1 and N = 40

$\Rightarrow LM + N = 36 + 40 = 76$

53. 1

Sol. $\lim\limits_{t \to x+1} \frac{t^2 f(x+1) - (x+1)^2 f(t)}{f(t) - f(x+1)} = 1\left(\frac{0}{0}\right)$

$\Rightarrow \lim\limits_{t \to x+1} \frac{2t f(x+1) - (x+1)^2 f'(t)}{f'(t)} = 1$

$\Rightarrow 2(x+1) f(x+1) - (x+1)^2 f'(x+1) = f'(x+1)$

$\Rightarrow 2(x+1).f(x+1) = \left[1 + (x+1)^2\right] f'(x+1)$

$\Rightarrow \frac{f'(x+1)}{f(x+1)} = \frac{2(x+1)}{1+(x+1)^2}$

Integrating, we get

$\ln(f(x+1)) = \ln\left(1 + (x+1)^2\right) + \ln c$

Put $x = -1$

$0 = 0 + \ln c$

$\Rightarrow c = 1$

$\Rightarrow f(x+1) = 1 + (x+1)^2$

So, $f(x) = 1 + x^2$

Hence, $\lim\limits_{x \to 1} \frac{\ln(f(x)) - \ln 2}{x-1} = \lim\limits_{x \to 1} \frac{\ln(1+x^2) - \ln 2}{(x-1)} = 1$

54. 2

Sol. $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 = 0, \alpha^5 = 1$

$1 + \alpha + \alpha^2 + \alpha^3 - \frac{2}{\alpha} = -\alpha^4 - \frac{2}{\alpha} = -3\alpha^4$

$\left|1 + \alpha + \alpha^2 + \alpha^3 - \frac{2}{\alpha}\right| = \left|-3\alpha^4\right| = 3$

$\therefore \log_{\sqrt{3}} 3 = 2$